

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **XCTest framework**-based automated test suites in **Swift** for enterprise applications, achieving **92% code coverage** and reducing manual testing effort by **65%** across iOS/macOS platform validations.
- Implemented **LLM-powered test generation** using **Claude API and Anthropic SDK** to automatically generate intelligent test cases from requirement specifications, reducing test authoring time by **40%** and improving defect detection accuracy by **28%**.
- Orchestrated end-to-end **GenAI and LLM integration** in test automation workflows, including prompt engineering, structured-JSON constraint guardrails, and automated defect analysis systems that prioritized critical failures with **94% precision**.
- Engineered **CI/CD pipeline integration** using **Jenkins, Git, and GitHub Actions** to execute comprehensive automated testing with detailed performance metrics, enabling continuous deployment across localization and release engineering workflows with zero-touch reporting.
- Performed **localization and internationalization testing** across SEPA, cross-border, and multi-currency payment flows in banking systems; validated **accessibility testing** standards and regulatory compliance (GDPR, data protection) in production release cycles.
- Developed **machine learning model training and fine-tuning** for intelligent test prioritization using **scikit-learn and pandas**, achieving **35% reduction** in test execution time while maintaining **99.2% reliability** in defect detection across Apple platform releases.

SKILLS

Automation Testing & Frameworks – XCTest framework, Swift programming, automated test suite design, test strategy development, comprehensive test automation.

GenAI & LLM Technologies – Claude API, Anthropic SDK, prompt engineering, LLM-powered test generation, intelligent defect analysis, constraint guardrails.

Machine Learning & Analytics – Machine learning model training, fine-tuning, scikit-learn, pandas, NumPy, supervised and unsupervised learning, model evaluation metrics.

CI/CD & DevOps – Jenkins pipelines, GitHub Actions, Git branching, continuous deployment, automated reporting, performance metrics analysis.

Programming & Data Technologies – Swift, Python, Java, SQL, REST APIs, data pipeline orchestration, ETL workflows.

Testing Specializations – Localization testing, internationalization testing, accessibility testing, performance metrics execution, defect analysis.

Platforms & Tools – iOS/macOS application testing, cloud infrastructure, Docker containerization, Linux command-line operations.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

LLM-Powered Test Automation Framework (HCL Technologies)

- Designed and implemented an **LLM-powered test generation system** leveraging **Claude API (Anthropic SDK)** to autonomously generate **XCTest framework**-based test cases from natural-language requirement specifications, reducing manual test authoring effort by **40%**.
- Engineered **prompt engineering strategies** with structured-JSON constraint guardrails to ensure generated test cases adhered to platform-specific testing standards for iOS/macOS applications; achieved **94% test case validity** on first generation.
- Integrated **intelligent defect analysis systems** powered by LLMs to classify and prioritize test failures in real-time; integrated with **Jenkins CI/CD pipelines** and **GitHub Actions** for automated test execution and continuous deployment reporting.
- Deployed framework across enterprise release engineering workflows, supporting **localization and internationalization testing** for global software releases; enabled zero-touch test execution with detailed performance metrics and **35% faster** test cycle times.

AI-Driven Resume Generation & Release Automation Platform (Personal Project)

- Built end-to-end **GenAI automation pipeline** orchestrating **Claude API (Anthropic SDK)** integration with **Python**-based job description ingestion, prompt engineering, and structured-JSON response generation for large-scale resume tailoring.
- Implemented **machine learning-based content optimization** using **scikit-learn** and **pandas** to analyze job descriptions and score resume alignment, achieving **92% precision** in skill-match detection.
- Engineered **CI/CD automation** using **GitHub Actions** for continuous testing and deployment to **Vercel**; integrated **reportlab PDF generation** for automated report generation and embedded **performance metrics tracking**.
- Demonstrated advanced **prompt engineering** with constraint guardrails, structured outputs, and LLM fine-tuning concepts directly applicable to intelligent test generation and defect analysis in automated testing environments.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)